

Lab 01 - Expected evidence

Everything below is what a learner should be able to show at the end of the lab. Node names, ages, UIDs, IP addresses and image digests will differ on your cluster; shape and key strings are what matters.

1. Cluster identity

```
$ kubectl version
Client Version: v1.37.0
Kustomize Version: v5.8.1
Server Version: v1.37.0
```

```
$ kubectl get nodes -o wide
NAME                                STATUS    ROLES    AGE   VERSION   INTERNAL-IP
EXTERNAL-IP  OS-IMAGE                KERNEL-VERSION   CONTAINER-RUNTIME
kcna-qa-20260905-control-plane  Ready    control-plane   47m   v1.37.0   172.18.0.2   <none>
Debian GNU/Linux 12 (bookworm)  6.10.14-linuxkit containerd://2.1.4
```

A single-node kind cluster shows exactly one node and it carries the control-plane role. On a managed cloud cluster you would see several nodes and the control plane would not be listed at all.

2. Namespace and guardrails

```
$ kubectl apply -f manifests/00-namespace.yaml
namespace/kcna-lab01 created

$ kubectl apply -f manifests/10-namespace-guardrails.yaml
resourcequota/depot-recon-quota created
limitrange/depot-recon-defaults created

$ kubectl -n kcna-lab01 get resourcequota depot-recon-quota
NAME                AGE   REQUEST                                LIMIT
depot-recon-quota   9s    count/configmaps: 0/10, pods: 0/10, requests.cpu: 0/1, limits.cpu: 0/2, limits.memory: 0/2Gi
requests.memory: 0/1Gi
```

3. API discovery

```
$ kubectl api-versions | head -n 8
admissionregistration.k8s.io/v1
apiextensions.k8s.io/v1
apiregistration.k8s.io/v1
apps/v1
authentication.k8s.io/v1
authorization.k8s.io/v1
autoscaling/v1
autoscaling/v2
```

```
$ kubectl api-resources --api-group='' --no-headers | head -n 6
bindings                true      Binding
configmaps              cm        true      ConfigMap
endpoints               ep        true      Endpoints
events                  ev        true      Event
limitranges             limits    true      LimitRange
namespaces              ns        false     Namespace
```

Column 3 is NAMESPACED. namespaces itself is false - namespaces are cluster-scoped, which is why kubectl get ns -n something is meaningless.

4. Baseline diff

```
$ kubectl api-resources --api-group='' --no-headers | awk '{print $1}' | sort > /tmp/live-core.txt
$ grep -v '^#' data/core-api-inventory.tsv | cut -f1 | sort > /tmp/baseline-core.txt
$ comm -23 /tmp/baseline-core.txt /tmp/live-core.txt
```

Empty output = every baseline resource is served by this cluster. That is the pass condition. The reverse direction shows what this cluster serves *beyond* the baseline:

```
$ comm -13 /tmp/baseline-core.txt /tmp/live-core.txt
bindings
endpoints
```

endpoints appears because v1 Endpoints is deprecated (v1.33) but still served.

5. Schema exploration

```
$ kubectl explain pod.spec.containers.resources
GROUP:
KIND:      Pod
VERSION:   v1

FIELD: resources <ResourceRequirements>

DESCRIPTION:
  Compute Resources required by this container. Cannot be updated.

FIELDS:
  claims      <[]ResourceClaim>
  limits      <map[string]Quantity>
  requests    <map[string]Quantity>
```

6. The Pod and the mounted dataset

```
$ kubectl -n kcna-lab01 get pod recon-shell
NAME          READY   STATUS    RESTARTS   AGE
recon-shell   1/1     Running   0           21s

$ kubectl -n kcna-lab01 logs recon-shell | tail -n 4
RC-011,state,What does the API server store for one object?,kubectl get pod recon-shell -n kcna-lab01 -o yaml,metadata.uid resourceVersion and status blocks
RC-012,events,What has the control plane recently done here?,kubectl get events -n kcna-lab01 -sort-by=.lastTimestamp,Scheduled Pulled Created Started
[recon-shell] checklist rows: 13
[recon-shell] holding open for kubectl exec
```

7. kubectl is an HTTP client

```
$ kubectl -n kcna-lab01 get pods --v=6
I0905 15:44:02.118374 62311 loader.go:395] Config loaded from file: /tmp/kcna-kind-20260905/
kubeconfig
I0905 15:44:02.146902 62311 round_tripper.go:553] GET https://127.0.0.1:52913/api/v1/
namespaces/kcna-lab01/pods?limit=500 200 OK in 21 milliseconds
NAME          READY   STATUS    RESTARTS   AGE
recon-shell   1/1     Running   0           64s
```

8. Verification script

```
$ bash verification/checks.sh

== 0. kubectl and cluster reachability
[PASS] kubectl can reach the API server

== 1. Namespace and namespaced policy objects
[PASS] namespace kcna-lab01 exists
[PASS] ResourceQuota depot-recon-quota exists
[PASS] LimitRange depot-recon-defaults exists

== 2. ConfigMap built from data/recon-checklist.csv
[PASS] ConfigMap recon-checklist exists
[PASS] ConfigMap carries key recon-checklist.csv
[PASS] ConfigMap content includes checklist row RC-010

== 3. recon-shell Pod
[PASS] Pod recon-shell phase is Running
[PASS] container 'shell' reports ready=true
[PASS] recon-shell logs report 13 lines (CSV header plus 12 checklist records) read from the
mounted CSV

== 4. Core API inventory matches data/core-api-inventory.tsv
[PASS] all 14 baseline core resources are served by this cluster

== 5. kubectl explain reaches the live OpenAPI schema
[PASS] kubectl explain pod.spec.containers.resources returns the 'requests' field

== 6. Verbose mode exposes the REST call
[PASS] --v=6 shows GET .../api/v1/namespaces/kcna-lab01/pods

Result: 14 passed, 0 failed
```